Rosco®

A LeeBoy Company

OPERATIONS, SERVICE AND PARTS MANUAL



RB50 BROOM

Manual No. 1017258-03

Rev. B

This manual applies to Serial Number 255027 and above.

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Disclaimer

All information, illustrations and specifications in this manual are based on the latest information available at the time of publishing. The illustrations used in this manual are intended as representative reference views only. Moreover, because of our continuous product improvement policy, we may modify information, illustrations and/or specifications to explain and/or exemplify a product, service or maintenance improvement. We reserve the right to make any change at any time without notice. ST Engineering LeeBoy, Inc., dba LeeBoy, ("LeeBoy"), is referred to as LeeBoy throughout this manual.

Title 40, Code of Federal Regulations (CFR) 1068

This product meets certified-emission requirements set by the EPA (Environmental Protection Agency), governed by Title 40 CFR 1068, which specifies actions that are prohibited by law and lists civil penalties for noncompliance. As part of those regulations, modification or rendering inoperative any emission-related component can subject you to government penalties (and void your warranty). Tampering with emission controls is in violation of federal law, and can result in civil penalties of up to $3,750 each day an engine or piece of equipment is operated in violation.

Please be aware that you are responsible for maintaining the machine and the certified emission engine installation. Failure to comply to comply could result in penalties as listed above and void the warranty on this engine and this machine.

For more information, visit: https://www.epa.gov/laws-regulations/regulations

California Proposition 65 Warning

Diesel engine exhaust and some of its constituents are known to the State of California to cause cancer, birth defects, and other reproductive harm.

Battery posts, terminals and other related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer and other reproductive harm. Wash hands after handling.

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NOTES

Thank you for purchasing the Rosco RB50 Broom. We wish you many years of safe and efficient operation of your machine.

READ THIS MANUAL PRIOR TO OPERATING the unit. This is an important part of the machine and should be kept in the dedicated storage container at all times. Though you may be familiar with similar equipment, you MUST read and understand this manual before operating this machine to help prevent injury or damage to the unit. If this manual becomes lost or damaged, contact your authorized Rosco LeeBoy dealer immediately to order a replacement (see **Contact Information** in **Section 2**).

This manual is intended as a guide for the safe and efficient use for proper operation and maintenance of the machine. This manual contains information that was available at the time of printing and are subject to change without notice.

Use this manual along with all related supplemental books, engine and transmission manuals included with your machine. Related Service Bulletins should be reviewed to provide information regarding some of the recent changes. Contact your authorized Rosco LeeBoy dealer for the latest available information.

Section 1 - Safety: Contains general and specific safety guidelines for product and safety label locations.

Section 2 - Information and Specifications: Contains warranty, contact information, machine specification tables, and machine dimensions.

Section 3 - Component Location: Contains overview of major component locations and functions.

Section 4 - Operation: Contains instructions for safe operation and information for optional equipment.

Section 5 - Maintenance: Contains routine maintenance procedures, mechanical adjustments, component replacement and troubleshooting charts for common problems and corrections. (For specific engine maintenance procedures, refer to the engine manufacturer manual.)

Section 6 - Schematics: Contains electrical and hydraulic schematics for product functionality.

Section 7 - Illustrated Parts List (IPL): Contains parts numbers and illustrations for serviceable components.

LeeBoy is proud to be ISO 9001 certified. The International Standards Organization (ISO) establishes guidelines to ensure that products and services are safe, reliable, and of good quality. ISO certifies companies who demonstrate compliance with all aspects of product safety, customer satisfaction, efficiency, environmental stewardship and social responsibility. Our teams work hard to deliver quality industrial machines that exceed customer expectations and we strive for continuous improvement in everything we do. The LeeBoy family of companies is committed to total quality management with a strong focus on meeting customer needs.

LeeBoy is also proud to be an accredited ANAB manufacturer, which is a certification process comprised of quality standards established by the American National Standards Institute (ANSI) and the American Society for Quality (ASQ). The ANSI-ASQ National Accreditation Board plays an important role in ensuring the safety and quality of goods and services, along with protecting the environment.

NOTES

Section 1

SAFETY

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NOTES

This manual provides important information to familiarize you with safe operating and maintenance procedures. Even though you may be familiar with similar equipment, you MUST read and understand this manual before operating the Rosco RB50 Broom and follow its instructions when operating the broom.

Safety is everyone's business and our top concern. Knowing the guidelines covered in this section will help ensure your safety, the safety of those around you, as well as proper broom operation.

Keep safety decals in good condition. If safety decals become missing or damaged, replace them immediately. Replacement safety decals are available from your authorized Rosco/LeeBoy dealer.

You can find more information about occupational health and safety in the industry on the internet. A few resources are listed below:

www.osha.gov

cdc.gov

www.safety.fhwa.dot.gov/

LOOK FOR THESE SYMBOLS THROUGHOUT THIS MANUAL. THESE ITEMS ARE EXTREMELY IMPORTANT FOR THE SAFETY OF YOU AND YOUR COWORKERS. READ AND UNDERSTAND THOROUGHLY. HEED THE WARNINGS AND FOLLOW THE INSTRUCTIONS.

⚠ DANGER Indicates a hazardous situation which, if not avoided, *will* result in death or serious injury.

⚠ WARNING Indicates a hazardous situation which, if not avoided, *could* result in death or serious injury.

⚠ CAUTION Indicates a hazardous situation which, if not avoided, *could* result in minor or moderate injury.

NOTICE **Indicates a situation which can cause damage to the equipment, personal property and/or the environment, or cause the machine to operate improperly.**

NOTE: Indicates a procedure, practice or condition that should be followed in order for the machine or component to function in the manner intended.

SAFETY PRECAUTIONS

⚠ CAUTION

The safety messages that follow have CAUTION level hazards.

Pre-Operation Hazard



Read and understand this manual before operating or servicing the engine to ensure that safe operating practices and maintenance procedures are followed.

- Never permit anyone to service or operate the machine without proper training.
- Safety signs and labels are additional reminders for safe operating and maintenance techniques.
- Contact your authorized Rosco/LeeBoy dealer for additional training.
- Make sure you are aware of all laws and regulations that are in effect for the location in which the broom is operated.
- Make sure you have all necessary licenses to operate the broom.

The safety messages that follow have CAUTION level hazards.

Poor Lighting Hazard



The work area must be well lit to ensure safe and proper operation.

- Ensure that the work area is adequately illuminated.
- Always install wire cages on portable safety lights.

Tool Hazard

Always use tools appropriate for the task at hand and use the correct size tool for loosening or tightening machine parts.

⚠ DANGER

The safety messages that follow have DANGER level hazards.

Power Lines Hazard

If your machine comes into contact with electric power lines, observe the following:

- Stay in the operators seat.
- Warn other workers to stay away and do not touch any control or any part of the machine.
- If contact can be broken, drive the machine away from the danger zone.
- If contact cannot be broken, stay in the operators seat until told that power is off.
- Failure to observe these directions could result in electrocution or death.

Electrocution Hazard

Disconnect the battery before welding anywhere on the machine.

Suffocation Hazard



Carbon monoxide poisoning is a serious condition that occurs as a result of improper ventilation.

- Never operate the internal combustion engine on this machine in an enclosed area with poor ventilation. Ensure proper ventilation to reduce risk of carbon monoxide poisoning or death.

Exhaust Hazard



All internal combustion engines create carbon monoxide gas during operation and special precautions are required to avoid carbon monoxide poisoning:

- Never block windows, vents or other means of ventilation.
- Always ensure that all connections are tightened to specifications after repair is made to the exhaust system.

⚠ WARNING

The safety messages that follow have WARNING level hazards.

Crush Hazard

Keep bystanders away from work area before and during operation.

Modification Hazard

Never modify the machine without the written consent of Rosco/LeeBoy . Any modification can affect safe operation and may cause personal injury or even death.

Exposure Hazard



Operators must be aware of their work environment and the equipment needed to work safely.

- Always wear personal protective equipment, including appropriate clothing, gloves, work shoes, and protection for eyes and ears, as required by the task at hand.

Explosion Hazard



While the engine is running or the battery is charging, hydrogen gas is being produced and can be easily ignited. Keep the area around the battery well-ventilated and keep sparks, open flame, and any other form of ignition out of the area.

- Always disconnect the negative (-) battery cable before servicing the broom.
- Do not start the engine by shorting the starter circuit or any other starting method not stated in this manual. Only use the starting procedure as described in this manual to start the engine.
- Never charge a frozen battery. Always slowly warm the battery to room temperature before charging.

Fire and Explosion Hazard

- Diesel fuel is flammable and explosive under certain conditions.
- Never use a shop rag to catch fuel.
- Wipe up all spills immediately.
- Never refuel with the engine running.
- Store any containers containing fuel in a well-ventilated area, away from any combustibles or sources of ignition.

Fire Hazard



When operating machinery there is a risk for fire. Always ensure appropriate safety equipment is available.

- Keep a charged fire extinguisher within reach when working in an environment where a fire may occur.
- Have all fire extinguishers checked periodically for proper operation and/or readiness.
- Always read and follow safety-related precautions found on containers of hazardous substances like parts cleaners, primers, sealants and sealant removers.
- Undersized wiring systems can cause electrical fires.

Entanglement/Sever Hazard



Verify there are no people, obstacles or other equipment near the machine before starting the engine. Sound the horn as a warning before starting the engine.



If the engine must be serviced while it is operating, remove all jewelry and tie back long hair before operating or servicing the machine.

- Keep hands, other body parts, and clothing away from moving/rotating parts.
- Always stop the engine before beginning service. Before maintenance, remove negative battery cable from battery post to ensure vehicle is not operated during maintenance.
- Verify that all guards and covers are properly attached before starting the engine. DO NOT start the engine if any guards or covers are not properly installed on the machine.

- If you must run the engine during maintenance procedures, make sure you have a helper to keep bystanders clear of the machine and make observations of moving parts as requested by the operator.
- Always turn the start switch to the OFF position after operation is complete and remove the key from the switch. Keep the key in your possession when the broom is not operating.
- Attach a “Do Not Operate” tag near the key switch while performing maintenance on the equipment.
- Never operate the engine while wearing a headset to listen to a radio or music because it will be difficult to hear the warning signals.
- Always start the engine and operate the controls while seated in the operators seat.

Alcohol and Drug Hazard



Never operate the machine while under the influence of alcohol, drugs, or when ill.

Piercing Hazard



High-pressure hydraulic fluid or fuel can penetrate your skin and result in serious injury. Avoid skin contact with high-pressure hydraulic fluid or diesel fuel spray caused by a hydraulic or fuel system leak such as a broken hydraulic hose or fuel injection line.

- If you are exposed to high-pressure hydraulic fluid or fuel spray, obtain prompt medical treatment.
- Never check for a hydraulic fluid or fuel leak with your hands. Always use a piece of wood or cardboard. Have your authorized LeeBoy dealer or distributor repair the damaged parts.

Flying Object Hazard



Always wear eye protection when cleaning the machine with compressed air or high-pressure water.

Dust, flying debris, compressed air, pressurized water or steam may cause eye injury.

Coolant Hazard



Coolant must be handled properly to ensure operator safety.

- Wear eye protection and rubber gloves when handling engine coolant.
- If contact with the eyes occurs, flush eyes with clean water for 15 minutes.
- If contact with skin occurs, wash immediately with soap and clean water.

Burn Hazard



Some of the machine’s surfaces become very hot during operation and shortly after shutdown.

- Keep hands and other body parts away from hot machine surfaces.
- Handle hot components with heat-resistant gloves.

NOTICE

The safety messages that follow have NOTICE level hazards.

Any part that is found defective as a result of inspection or any part whose measured value does not satisfy the standard or limit must be replaced.

Always tighten components to the specified torque. Loose parts can cause damage to the machine or cause it to operate improperly.

Only use replacement parts approved by Rosco/ LeeBoy. Other replacement parts may affect warranty coverage.



Follow the guidelines of the EPA or other governmental agencies for the proper disposal of hazardous materials such as engine oil, diesel fuel, and engine coolant. Consult the local authorities or reclamation facility.

Dispose of hazardous materials in accordance with all applicable laws and regulations. Never dispose of hazardous materials by dumping them into a sewer, on the ground, or into groundwater or waterways.

Clean all accumulated dirt and debris away from the body of the broom and its components before you inspect the broom or perform preventive maintenance procedures or repairs. Operating a machine with accumulated dirt and debris will cause premature wear of components.

If any alert indicator illuminates during operation, stop the engine immediately. Determine the cause and repair the problem before continuing to operate.

MACHINE PRECAUTIONS

Hot Material Precautions

- Allow machine to cool before repairing or maintaining working components.
- DO NOT remove radiator cap, drain plugs, service grease fittings, or pressure taps while engine is hot. Add coolant to the radiator and perform other services only when the engine is stopped and fully cooled.

Hydraulic Systems Precautions

- Ensure all components are in good working condition. Replace any worn, cut, abraded, flattened or crimped hoses and metal lines.
- DO NOT attempt makeshift repairs using tape, clamps or cements. The hydraulic system operates under extremely high pressure and such repairs could cause serious injury.
- Wear proper hand and eye protection when checking for a high pressure leak. Use a piece of wood or cardboard as a back stop to isolate and identify leaks.

⚠WARNING Hydraulic oil under pressure can cause serious personal injury. Check for oil leaks with a piece of cardboard. DO NOT expose hands to possible high-pressure oil. Turn off engine before attempting to tighten oil lines and fittings.

- Escaping pressurized hydraulic fluid has force sufficient to penetrate the skin, which could cause serious personal injury. Ensure all pressure is relieved before disconnecting line, hoses or valves.
- If injury from concentrated high pressure steam or hydraulic fluid occurs, seek medical attention immediately. Injuries resulting from hydraulic fluid penetrating the skin's surface can result in serious infections or toxic reactions.

Refueling Precautions

- Do NOT overfill the fuel tank as overflow creates a fire hazard when spilled on hot components.
- DO NOT smoke when refueling and never refuel when the engine is running. Fuel is highly flammable and should be handled with care. Death or serious injury can occur due to explosion and/or fire.
- DO NOT fill tank to capacity. Allow room for expansion to reduce the risk of fuel expanding and spilling from the tank.
- Tighten fuel cap securely. Should fuel cap be lost, replace it with an original manufacturer's approved cap. Pressurization of the tank may result from use of non-approved cap.
- Prevent fires by keeping the machine clean of accumulated debris, grease, and spilled fuel.
- Use ultra-low sulfur diesel fuel (ULSD) only.

Battery Precautions

- Keep all sparks and flames away from batteries, as gas given off by electrolytes is explosive.
- Acid propelled by an explosion can cause blindness if it comes in contact with eyes. Always wear safety glasses when working near batteries.
- If you come in contact with battery electrolyte solution, wash off immediately. Chemicals can cause burns.
- Always disconnect the battery ground cable before working on the electrical system to avoid injury from spark or short circuit. Electrical shock and burns can occur.
- To avoid electrolyte loss, DO NOT tip batteries more than 45 degrees.
- Use jumper cables ONLY as directed in Section 5 (Maintenance). Improper use can result in battery explosion or unexpected machine motion.

Starting and Stopping Precautions

- Check all around the machine to ensure there are no people working on the machine or in its path before starting. DO NOT start until area is clear. Death or serious injury can occur to bystanders from being crushed under a moving machine.

⚠WARNING **DO NOT operate the engine in an enclosed area without proper ventilation. Exhaust gasses are odorless and deadly.**

Parking Precautions

- Park machine on level ground whenever possible and apply the parking brake.
- Before leaving operator's station:
  1. Turn off all accessories.
  2. Set parking brake.
  3. Shut off engine.
- Always remove ignition key when leaving the machine parked or unattended.

Operating Precautions

- Always comply with local regulations regarding industrial equipment on public roads and highways.
- Know and use the hand signals required for a particular job. Know who has the responsibility for signaling.
- Make sure that all lights and reflectors comply with state and local regulations. Ensure that they are clean, in good working order, and can be seen clearly by all traffic.
- DO NOT ride on attachments.
- Check all gauges and warning instruments for proper operation. If malfunctions are found, shut down the machine and report the problem for resolution. If the failure causes loss of steering control, loss of brake control, or loss of engine power, stop motion as quickly as possible. Apply parking brake and turn off the machine. Keep the machine securely parked until the failure is corrected or the machine can be safely transported.

- Drive the machine with care. Make sure speed is compatible with conditions. Use caution on rough ground, slopes, in traffic zones and while turning.
- Be alert for hazards and obstructions such as ditches, trees, cliffs, overhead power lines, and areas where there is danger of a slide.
- Be aware of and understand the job site traffic flow patterns.
- Obey flagmen, road signs, and signals.
- Watch for bystanders. Never allow anyone to be under the machine during operation. Never allow anyone to reach into the machine during use.
- Operator must know how to use signaling devices and understand which circumstances require use of each signal. Use tail lights, slow moving vehicle signs, and warning beacon as needed when traveling on public roads. It is recommended that you provide an escort on the road.
- DO NOT tow the machine, except to remove from road or load on a trailer for transport.

Storage Precautions

- Store machine in an area away from human activity.
- DO NOT permit children to play on or around the machine. Serious injury or death can occur from improper/unauthorized use of the machine.
- Make sure the unit is stored on a surface that is firm, level, and free of debris.
- Store the machine inside a building or cover securely with a weatherproof tarpaulin.

Maintenance Precautions

- DO NOT attempt repairs unless trained to do so. Refer to manuals and experienced repair personnel for help.
- Before working on the machine, securely block the machine and any components that may fall. Block any working components to prevent unexpected movement while repairs are being made.
- Always wear safety glasses and other required safety equipment when servicing or making repairs.
- Disconnect battery before working on the electrical system.
- Avoid lubrication or mechanical adjustments while the machine is in motion and engine is operating.
- If lubrication or mechanical adjustment is necessary, use extreme caution.
- Never make repairs on pressurized components such as fluid lines, the gas system, or mechanical items until the pressure has been relieved.
- When servicing or replacing hardened pins, use a brass drift or other suitable material between the hammer and pin.
- Keep brake and steering systems in good operating condition.

Tire Precautions

⚠WARNING DO NOT mount or demount tires without proper training. A violent explosion can occur.

Follow all procedures and safety instructions. Charts containing mounting and demounting instructions are available from OSHA at www.osha.gov/pls/publications.

NOTICE **Contact your authorized dealer or a qualified tire dealer or repair service to perform required tire maintenance.**

- Clean rims and repaint to stop corrosion and facilitate checking air level and tire mounting. Be careful to clean all dirt and rust from the lock ring and gutter. This is important to secure lock ring in its the proper position.
- Check rim components periodically for cracks. Replace all cracked, badly worn, severely rusted, and otherwise damaged components with new parts of the same size and type. If you are not sure about proper mating of tire and wheel parts, consult your authorized dealer or a wheel and rim expert.

⚠WARNING DO NOT attempt to rework, weld, heat or braze any rim components that are cracked, broken or damaged. Replace damaged components with new parts of the same size and type. Mixing parts of one type with those of another is potentially dangerous.

⚠CAUTION DO NOT reinflate a tire that has been run flat without first inspecting the tire, tube, flap, rim, and wheel assembly. Double-check the side ring and O-ring for damage, ensuring they are secured in the gutter before inflation.

⚠WARNING Mismatched rim parts are dangerous and could cause SEVERE injury.

- Remove valve core to exhaust all air from tire. ALWAYS exhaust all air from tire prior to removing rim components or wheel components such as nuts or rim clamps. Check the valve stem by running a piece of wire through the stem to make sure it is not plugged.
- DO NOT attempt to demount a tire unless you have the proper equipment and experience to do the job.
- DO NOT try to seat rings or other components by hammering unless tire is fully deflated. Double-check all components prior to inflation.
- DO NOT inflate a tire before all components are properly in place. Recheck components for proper assembly. If improperly assembled, deflate and correct.
- Never hammer on a tire/rim assembly that is not fully deflated. If assembly is proper at 10 psi (68,947.572 Pa), continue to inflate to fully seat the tire beads. Then completely deflate the tire to prevent overly stretching the tube. Reinflate to recommended operating pressure.
- NEVER sit on or stand in front of a tire/rim assembly during inflation. Use a clip-on chuck and make sure inflation hose is long enough permit the person inflating the tire to stand to the side of the tire.
- DO NOT hammer components with steel hammers. Use rubber, lead, plastic, or brass-faced mallets to tap components together.
- DO NOT try to drive an assembled or partially-assembled tire or rim over a cast spoke wheel by hammering. DEFLATE and examine to determine the reason for the improper fit. Look for distortion or components that are not properly locked or sealed.
- Block the tire and wheel on opposite sides of the vehicle before you place the jack in position. Always put hardwood blocks under the jack and crib up the vehicle in case the jack slips.
- DO NOT inflate tires beyond the maximum recommended inflation pressure.
- NEVER operate a vehicle on a single tire of a dual assembly. The carrying capacity of the single tire and rim is dangerously exceeded and operating a vehicle in this manner can result in damage to the rim and tire.

SAFETY DECALS

If your machine is repainted, it is extremely important that you replace all the CAUTION, WARNING and DANGER safety decals as shown in the following graphics. For additional help, refer to the parts listing in **Section 7** and contact your authorized LeeBoy dealer to order a replacement kit.

NOTE: It is the responsibility of the owner and operator to make sure that all safety labels are readable and located on the machine as designated by LeeBoy.

Safety Decal Locations

